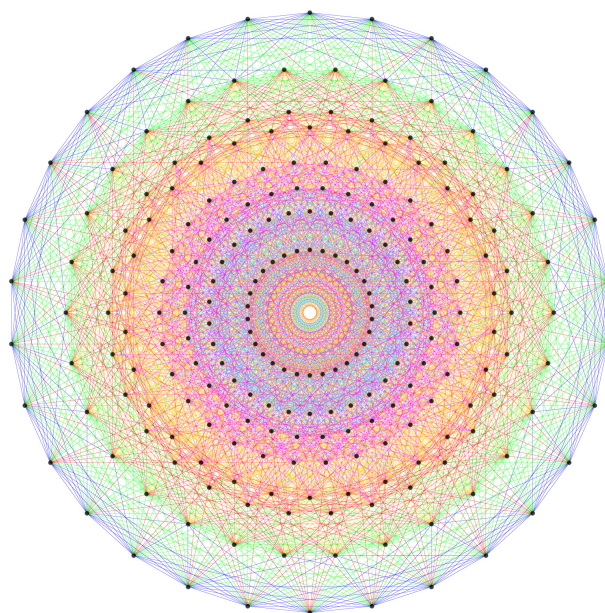


Representation Theory

Maschke theorem: Brief Exposition

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Introduction

This exposition aims at exploring the basic foundations of representation theory of finite groups. The main outcome is to explore Maschke's theorem that provides a sufficient condition for when a group algebra is semisimple. We also explore Artin-Wedderburn theorem that characterizes the structure of the semisimple algebras irreducible and finite dimensional representations. The exposition assumes basic knowledge in group, ring and field theory and linear algebra

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Chapter 1

Linear Algebra

We begin by defining what is an Algebra. This is the main structure we will be using throughout this paper. To give you an intuition first, an Algebra is basically a set with a structure of both a ring and a vector space.

1.1 Algebras over a Field

Definition 1.1 (k -Algebra)

Let k be any field. We say that a k -vector space A is a k -algebra if A is equipped with an associative bilinear operation (product)

$$\times : A \times A \rightarrow A$$

satisfying the following axioms of bilinearity. For every $x, y, z \in A$ and $a, b \in k$:

1. $z \times (x + y) = z \times x + z \times y$
2. $(x + y) \times z = x \times z + y \times z$
3. $(a \cdot x) \times (b \cdot y) = ab \cdot (x \times y)$

where $\cdot$ is the vector space scalar multiplication.

Another way to define a k -algebra is to first define it as a ring equipped with scalar multiplication operation by embedding the field k inside A .

Definition 1.2 (k -Algebra)

Let k be a field. A k -algebra A is a ring with an injective ring homomorphism $k \rightarrow A$, where for any $\lambda \in k$ and $a \in A$ we have that

$$\lambda \cdot a = a \cdot \lambda$$

i.e. multiplication by elements in the image of k in A is commutative.

Remarks:-

1. Observe that we can define the vector space structure on A in the definition above by defining the scalar product λa to be the product of the image of λ in A times a to get $\lambda \cdot a$ for all $\lambda \in k$ and $a \in A$. That is

$$\lambda a = \lambda \cdot a$$

2. An algebra product operation might not be commutative. If so we just say a commutative algebra.
3. We may or may not include 1 as a unity in our definition of an algebra but here we do. And the operation is associative.

Definition 1.3 (k -Algebra Homomorphism)

Let A, B be k -algebras. A function $\varphi : A \rightarrow B$ is a k -algebra homomorphism

if it is a linear map of vector spaces and for any $a \in A$ and $b \in B$ we have that

$$\varphi(ab) = \varphi(a)\varphi(b)$$

1.2 Direct Sum and Direct Product

Now, we get to differentiate between two important notions in linear algebra: Internal Direct Sum and External Direct Product. Sometimes for simplicity we just say Direct Sum and Direct Product.

Definition 1.4 (Internal Direct Sum)

Let V be a vector space. We say V is an internal direct sum of two subspaces $U, W \subseteq V$ iff every vector $v \in V$ could be written uniquely as a sum of two vectors u, w where $u \in U$, and $w \in W$.

That is

$$V = U + W.$$

there are unique vectors $u \in U$ and $w \in W$ s.t.

$$v = u + w.$$

The above definition tells us exactly when a sum of two subspaces $U + W$ of a given vector space V is a direct sum, and in this case we call V an internal direct sum since the two spaces U and W are subspaces. Next, we define an External Direct Product of two separate given spaces.

Definition 1.5 (External Direct Product)

Let K, L be two vector spaces. We define the *External Direct Product* of K and L to be the vector space

$$K \times L := \{(k, l) \mid k \in K, l \in L\}$$

with operations of component wise addition and scalar multiplication.

The next theorem provides a complete characterization for when an External Direct Product of two subspaces and their sum are exactly the same, giving also an equivalent condition for when a sum is an internal direct sum.

Theorem 1.6

Let U, W be subspaces of a vector space V . **TFAE**

1. The map

$$\pi : U \times W \rightarrow U + W$$

given by

$$(u, w) \mapsto u + w$$

is an isomorphism.

2. $U \cap W = \{0\}$.
3. Any vector $x \in U + W$ could be written uniquely in the form

$$u + w$$

with $u \in U$ and $w \in W$.

Proof. (1) $\Rightarrow$ (2) Suppose for the sake of contradiction that there exist a non-zero

vector

$$v \in U \cap W.$$

So $v \in U$ and $v \in W$. Hence via the isomorphism π , the element $(v, 0) \in U \times W$ maps to $v + 0 = v$ and $(v, 0) \in U \times W$ maps to $0 + v = v$ violating injectivity of π .

(2) $\Rightarrow$ (3) Suppose that there is $x \in U + W$ such that x has two different representations

$$x = a + b = c + d.$$

where $a, c \in U$ and $b, d \in W$ Then

$$a + b = c + d$$

$$\Rightarrow a - c = d - b.$$

Since U, W are subspaces that is closed under addition we get that $a - c \in U$ and $b - d \in W$. Since $a - c = d - b$, therefore,

$$a - c \in U \cap W.$$

which implies that $a - c = d - b = 0$. Hence, $a = c$ and $b = d$ as desired.

(3) $\Rightarrow$ (1) Since any vector $x \in U + W$ could be written as $u + w$, where $u \in U$ and $w \in W$ this implies that π is surjective as a preimage for any element $x = u + w$ is (u, w) . By uniqueness of this representation of elements in $U + W$ we get that the preimage of an element $x = u + w$ is unique. ■

So, if we have a vector space that is a sum of two subspaces

$$U + W = V$$

and satisfies one of the conditions above we say V is an internal direct sum of U, W

and we write

$$V = U \oplus W.$$

Moreover, we can see that the two notions of internal direct sum and external direct product coincide when K, L are subspaces of some vector space V and V is an external direct sum of K and L as proved the direct sum & product are isomorphic.

Definition 1.7 (Direct sum of Algebras)

Let A, B be two k -algebras. We define the direct sum $A \oplus B$ to be the k -algebra whose elements are pairs (a, b) where $a \in A$ and $b \in B$, with component wise operations.

Chapter 2

Basics of Representation Theory

2.1 Introduction and Motivation

The main goal of representation theory is to understand abstract algebraic structures—specifically groups and associative algebras—by concretely realizing their elements as linear transformations of vector spaces. By transforming non-linear group multiplication into the well-understood domain of linear algebra and matrix arithmetic, one converts difficult group-theoretic questions into tractable eigenvalue problems, trace computations, and spectral decompositions.

First we define what is a representation? then we move to defining a homomorphism and isomorphism of representations. Moreover, we introduce the definition of an important structure that we will be using extensively throughout this exposition which is the group algebra $k[G]$, then we will move to working with simple representations/modules. In between we show how can modules and representations be seen as the same thing.

2.2 Basic Definitions

Definition 2.1 (Representation of an Algebra)

Let A be a k -algebra, and V be a k -vector space. A *representation* of A is a pair (V, ρ) where $\rho : A \rightarrow \text{End}(V)$ is a k -algebra homomorphism.

Definition 2.2 (Subrepresentation)

Let (V, ρ) be a representation of any k -algebra A . We say that a subspace $W \subseteq V$ is a subrepresentation iff elements of W commute with the representation ρ or that W is invariant under the action of A . That is for any $a \in A$ and any $w \in W$ we have that

$$\rho(a) \cdot w \in W$$

Definition 2.3 (Intertwining Operator)

Let (V, ρ_1) and (W, ρ_2) be any two representations of a k -algebra A , an intertwining operator $T : V \rightarrow W$ is a linear map that commutes with the action of A , that is

$$T(\rho_1(a)(v)) = \rho_2(a)(T(v))$$

We may also say that T is a homomorphism of representations. If T is invertible, then it is an isomorphism of representations

2.2.1 Equivalence of Representations and Modules

Now, how are representations of any k -algebra A could be seen as A -modules?

We basically define the action of A on V by the action of the image of A in $\text{End}(V)$ as linear transformations. For the other direction, Let M be an A -module. Then M is an abelian group. We define scalar multiplication via the action of A , as we mentioned earlier A has a copy of the field k inside it, hence: for any $\lambda \in k$ and

any $m \in M$ we have that

$$\lambda \cdot m = \lambda m$$

where $\cdot$ is the action of A on M . It remains to prove that the action of A on M as a module defines group algebra homomorphism from A to $\text{End}(M)$.

Proof. Fix any $a \in A$, consider the map

$$\varphi(a) : M \rightarrow M$$

given by $m \mapsto a \cdot m$. By the axioms of a module action we get

- $\varphi(a)(m + n) = a \cdot (m + n) = a \cdot m + a \cdot n$
- $\varphi(a)(\lambda m) = a \cdot (\lambda m) = (a\lambda) \cdot m = (\lambda a) \cdot (m) = \lambda(\varphi(a)(m))$

the last equality is because we defined the image of the field k in A to be central. So, we proved that $\varphi(a)$ is a linear map for any $a \in A$. Now, we define the group algebra homomorphism

$$\rho : A \rightarrow \text{End}(M)$$

given by $a \mapsto \varphi(a)$. To see that it is an algebra homomorphism, first we use the linearity of $\varphi(a)$

- $\rho(a + b) = \varphi(a + b) = \varphi(a) + \varphi(b) = \rho(a) + \rho(b)$
- $\rho(\lambda a) = \varphi(\lambda a) = \lambda \varphi(a) = \lambda \rho(a)$

since M is an A module, then by associativity of the action and the definition of $\varphi(a)$ for any $a \in A$, we get that

- $\rho(ab) = \varphi(ab) = \varphi(a)\varphi(b) = \rho(a)\rho(b)$

■

2.3 The Group Algebra

Now, we move to a special type of algebra called the Group algebra $k[G]$, that will help us study the representations of a finite group G .

Definition 2.4 (The Group Algebra $k[G]$)

Let G be a finite group and k a field. The *group algebra* $k[G]$ is the k -vector space with basis given by the elements of G :

$$k[G] = \left\{ \sum_{g \in G} a_g g \mid a_g \in k \right\}.$$

The multiplication on $k[G]$ is defined by extending the group multiplication linearly:

$$\left(\sum_{g \in G} a_g g \right) \left(\sum_{h \in G} b_h h \right) = \sum_{g, h \in G} a_g b_h (gh) = \sum_{x \in G} \left(\sum_{gh=x} a_g b_h \right) x.$$

The vector space dimension of $k[G]$ over k is precisely the order of the group:

$$\dim_k(k[G]) = |G|.$$

Before we define a representation of a group, we define the general case of a representation of a group algebra.

Definition 2.5 (Representation of a group algebra $k[G]$)

Let G be a group. Let $k[G]$ be any group algebra over a field k . A representation of $k[G]$ is a pair (V, ρ) , where V is a k -vector space and

$$\rho : A \rightarrow \text{End}(V)$$

is a group algebra homomorphism.

The previous definition just says that we are trying to represent elements of an algebra with linear transformation or matrices.

2.4 Representation of a Group

We are done discussing the general case. But what about the definition of a group representation? There are two ways to view this definition. One way is to see it as a restriction of the definition of an algebra representation on the basis elements which are basically the elements of the group G . The other one, which I find more interesting is to see it as an extension of a group action.

First, recall the definition of a group action on an arbitrary set S .

Definition 2.6 (Group Action)

Let G be a multiplicative group, and S to be any set. We define a group action of G on S to be a function

$$\cdot : G \times S \rightarrow S$$

denoted by $g \cdot x$ for all $g \in G$ and $x \in S$, satisfying the following axioms:

1. $e \cdot x = x \quad \forall x \in S$
2. $g \cdot (h \cdot x) = gh \cdot x$

Using the previous two axioms of a group action, one can see that it actually induces a group homomorphism $\alpha : G \rightarrow \text{Sym}(S)$, where $\text{Sym}(S)$ is the group of all permutations on S with function composition.

Proof. Let $\cdot : G \times S \rightarrow S$ be a group action. Consider the map

$$\alpha : G \rightarrow \text{Sym}(S)$$

where $\alpha(g)$ is a bijection on S such that

$$\alpha(g)(x) = g \cdot x$$

for all $g \in G$ and $x \in S$. We claim that α is a group homomorphism. By the axioms of the group action, for any $g, h \in G$:

$$\alpha(gh)(x) = gh \cdot x = g \cdot (h \cdot x) = \alpha(g)(\alpha(h)(x)) = \alpha(g) \circ \alpha(h)(x)$$

■

Now, we repeat the same process but using a group action of a group G on a vector space V instead of an arbitrary set S requiring an additional property for the group action which is to be *linear*.

Definition 2.7 (Linear Group Action)

Let V be a k -vector space, and G be any group. Define the group action of G on V as the function

$$\cdot : G \times V \rightarrow V$$

that satisfies the following axioms:

1. $e \cdot v = v \quad \forall v \in V$
2. $g \cdot (h \cdot v) = gh \cdot v$
3. $g \cdot (v + w) = g \cdot v + g \cdot w$
4. $g \cdot (\lambda v) = \lambda(g \cdot v)$

This definition is in fact equivalent to having a group homomorphism $\varphi : G \rightarrow \text{GL}(V)$, exactly in the same way a regular group action gives rise to a homomorphism of groups from G to the symmetric group on an arbitrary set S . This pair of the homomorphism φ along with the vector space V are called a *representation* of the group G .

Definition 2.8 (Representation of a Group)

Let G be a group, and let V be a vector space over a field k . A *representation* of G is a pair (V, φ) where $\varphi : G \rightarrow \text{GL}(V)$ is a group homomorphism given by the linear action of G on V .

Definition 2.9 (G -Homomorphism)

Let (ρ_1, V_1) and (ρ_2, V_2) be two representations of a group G over k . A linear map $T : V_1 \rightarrow V_2$ is called an *intertwining operator* (or *G -homomorphism*) if it commutes with the G -action:

$$T \circ \rho_1(g) = \rho_2(g) \circ T \quad \text{for all } g \in G.$$

If T is an invertible linear isomorphism, we say that ρ_1 and ρ_2 are *equivalent* (or *isomorphic*), denoted $\rho_1 \cong \rho_2$.

Definition 2.10 (Subrepresentation)

Let $\rho : G \rightarrow \text{GL}(V)$ be a representation. A vector subspace $W \subseteq V$ is called a *G -invariant subspace* (or *subrepresentation*) if

$$\rho(g)w \in W \quad \text{for all } g \in G \text{ and all } w \in W.$$

Definition 2.11 (Irreducible representation)

We say a representation (V, ρ) is irreducible if V has no proper non-zero subrepresentations.

Definition 2.12 (Indecomposable representation)

We say a representation (V, ρ) is indecomposable if there doesn't exist a pair of subrepresentations U, W such that $V = U \oplus W$.

Note. It is essential to distinguish between *irreducibility* and *indecomposability*:

- A representation V is *irreducible* if it admits no proper non-zero subrepresentation.
- A representation V is *indecomposable* if it cannot be written as a direct sum of two non-zero subrepresentations, i.e., $V \neq W_1 \oplus W_2$ with $W_1, W_2 \neq \{0\}$.

Every irreducible representation is indecomposable, but the converse does not hold in general fields! As we shall see in Section 3.7, when the characteristic of the field divides the group order, there exist indecomposable representations that are strictly reducible.

Chapter 3

Semi-Simple Algebras

3.1 Section Title

3.2 Definitions

Now, as we are done talking about the basics of representation theory, we get to the main goal of this exposition: introducing Semisimple algebras. The main claim we aim to prove in this chapter is that all semisimple algebras are product of matrix algebras. This characterization shows that semisimple algebras are the easiest to work with.

3.3 The Group Algebra and its Modules

A pivotal conceptual leap in modern representation theory is the realization that linear representations of a group G are in one-to-one correspondence with modules over an associative k -algebra, known as the *group algebra* $k[G]$. This bridge allows us to apply the full machinery of ring and module theory to finite group representations.

Definition 3.1 (Finite-Dimensional Associative Algebra)

A *finite-dimensional algebra* A over a field k is a k -vector space equipped with a bilinear associative multiplication

$$A \times A \longrightarrow A, \quad (a, b) \longmapsto a \cdot b$$

possessing a multiplicative identity $1_A \in A$. Bilinearity means that for all $a, b, c \in A$ and $\lambda \in k$:

$$(a + b)c = ac + bc, \quad a(b + c) = ab + ac, \quad (\lambda a)b = a(\lambda b) = \lambda(ab).$$

Theorem 3.2 (Equivalence of Representations and Modules)

Let G be a finite group and k a field. There is a canonical one-to-one correspondence between:

1. Linear representations of G over k , i.e., group homomorphisms $\rho: G \rightarrow \text{GL}(V)$;
2. Left $k[G]$ -modules $(V, \cdot)$, where the action is given by extending ρ linearly:

$$\left(\sum_{g \in G} a_g g \right) \cdot v = \sum_{g \in G} a_g \rho(g)v.$$

Under this correspondence:

- Subrepresentations of V correspond precisely to $k[G]$ -submodules of V .
- Irreducible representations of G correspond to simple $k[G]$ -modules.
- Intertwining operators between representations correspond to $k[G]$ -module homomorphisms.

Definition 3.3 (The Regular Representation)

The *left regular representation* of G is the vector space $V = k[G]$ equipped with the natural action of G on itself by left multiplication:

$$\rho_{\text{reg}}(g) \left(\sum_{h \in G} a_h h \right) = \sum_{h \in G} a_h (gh).$$

Viewing $k[G]$ as a left $k[G]$ -module acting on itself via ring multiplication, this is the free rank-one module ${}_A A$.

3.4 Semisimplicity and the Radical

To understand the structure of representations, we classify modules that decompose cleanly into fundamental irreducible building blocks.

Definition 3.4 (Simple and Semisimple Modules)

Let A be an algebra over k .

1. An A -module V is *simple* (or *irreducible*) if $V \neq \{0\}$ and its only submodules are $\{0\}$ and V .
2. An A -module V is *semisimple* (or *completely reducible*) if it is isomorphic to a direct sum of simple A -modules:

$$V \cong \bigoplus_{i \in I} V_i, \quad \text{where each } V_i \text{ is simple.}$$

3. An algebra A is called a *semisimple algebra* if A , regarded as a left module over itself (denoted ${}_A A$), is a semisimple left A -module.

Lemma 3.5 (Schur's Lemma)

Let A be an algebra over an algebraically closed field k .

1. If V_1 and V_2 are simple A -modules, then any A -module homomorphism $f: V_1 \rightarrow V_2$ is either zero or an isomorphism.
2. If V is a finite-dimensional simple A -module over k , then every A -module endomorphism is a scalar multiple of the identity:

$$\text{End}_A(V) \cong k \cdot \text{id}_V \cong k.$$

Proof. For (1), the kernel $\ker(f) \subseteq V_1$ is a submodule of V_1 . Since V_1 is simple, either $\ker(f) = V_1$ (implying $f = 0$) or $\ker(f) = \{0\}$ (implying f is injective). Similarly, the image $\text{im}(f) \subseteq V_2$ is a submodule of V_2 . Since V_2 is simple, either $\text{im}(f) = \{0\}$ (implying $f = 0$) or $\text{im}(f) = V_2$ (implying f is surjective). Thus, if $f \neq 0$, f is bijective, hence an isomorphism.

For (2), let $f \in \text{End}_A(V)$. Since V is a finite-dimensional vector space over the algebraically closed field k , the characteristic polynomial of f has a root $\lambda \in k$. Therefore, $f - \lambda \text{id}_V$ is not invertible, which means its kernel is non-trivial:

$$\ker(f - \lambda \text{id}_V) \neq \{0\}.$$

Since f and id_V are A -module homomorphisms, $f - \lambda \text{id}_V \in \text{End}_A(V)$. By part (1), since it is not an isomorphism, it must be the zero map. Hence $f - \lambda \text{id}_V = 0$, which proves $f = \lambda \text{id}_V$.

■

Definition 3.6 (The Radical)

Let A be a finite-dimensional algebra over k . The *Jacobson radical* of A , denoted $\text{Rad}(A)$, is the intersection of the annihilators of all simple left A -modules:

$$\text{Rad}(A) = \{a \in A \mid a \cdot v = 0 \text{ for all } v \in V \text{ and all simple left } A\text{-modules } V\}.$$

Equivalently, if $\{V_1, \dots, V_r\}$ is a complete set of representatives of isomorphism classes of simple A -modules with representation maps $\rho_i: A \rightarrow \text{End}_k(V_i)$, then

$$\text{Rad}(A) = \bigcap_{i=1}^r \ker(\rho_i).$$

Theorem 3.7 (Semisimplicity Criterion via Radical)

A finite-dimensional algebra A is semisimple if and only if its Jacobson radical is trivial:

$$\text{Rad}(A) = \{0\}.$$

Note. The endomorphism algebra of an algebra A acting on itself as a left module is canonically anti-isomorphic to A :

$$\text{End}_A({}_A A) \cong A^{\text{op}}.$$

To verify this, notice that any left A -linear map $\phi: A \rightarrow A$ is uniquely determined by its value on the unit 1_A :

$$\phi(a) = \phi(a \cdot 1_A) = a \cdot \phi(1_A).$$

Letting $b = \phi(1_A)$, ϕ is given by right multiplication by b : $\phi(a) = ab$. Then for two maps $\phi_b, \phi_c \in \text{End}_A(A)$, their composition satisfies:

$$(\phi_b \circ \phi_c)(1_A) = \phi_b(c) = cb,$$

which reverses the order of multiplication, yielding the opposite algebra A^{op} .

3.5 The Density Theorem and Artin-Wedderburn Theorem

We now address one of the deepest results from the notes: the Jacobson Density Theorem, which shows that the representation map $\bigoplus_{i=1}^r \rho_i: A \rightarrow \bigoplus_{i=1}^r \text{End}_k(V_i)$ is surjective.

Lemma 3.8 ()

Let A be an algebra over an algebraically closed field k , and let V be a finite-dimensional simple A -module. If $\{v_1, v_2, \dots, v_n\} \subset V$ is any linearly independent set of vectors, then for any arbitrary vectors $w_1, w_2, \dots, w_n \in V$, there exists an element $a \in A$ such that

$$a \cdot v_i = w_i \quad \text{for all } 1 \leq i \leq n.$$

Proof. Consider the direct sum $V^{\oplus n} = \underbrace{V \oplus \dots \oplus V}_{n \text{ times}}$. We define an A -module homomorphism

$$\Psi: A \longrightarrow V^{\oplus n}, \quad a \longmapsto (av_1, av_2, \dots, av_n).$$

Its image $U = \text{im}(\Psi) \subseteq V^{\oplus n}$ is an A -submodule of $V^{\oplus n}$. We must show that $U = V^{\oplus n}$, for if $U = V^{\oplus n}$, then $(w_1, \dots, w_n) \in U$, meaning there exists $a \in A$ with $\Psi(a) = (w_1, \dots, w_n)$, i.e., $av_i = w_i$ for all i .

Since V is a simple A -module, the direct sum $V^{\oplus n}$ is a semisimple A -module. By general module theory, every submodule of a semisimple module is a direct summand and isomorphic to a direct sum of a subset of the simple summands. Hence, there exists an integer $r \leq n$ such that

$$U \cong V^{\oplus r}.$$

We claim that $r = n$. Suppose for contradiction that $r < n$.

The inclusion map $\phi: U \hookrightarrow V^{\oplus n}$ is an A -module homomorphism from $V^{\oplus r}$ to $V^{\oplus n}$. By Schur's Lemma, since $\text{End}_A(V) \cong k$, any A -module homomorphism between $V^{\oplus r}$ and $V^{\oplus n}$ is represented by an $r \times n$ matrix $X \in \text{Mat}_{r \times n}(k)$:

$$\phi(u_1, \dots, u_r) = (u_1, \dots, u_r)X.$$

Since $r < n$, the rank of X is at most $r < n$. In particular, the column rank of X is strictly less than n , which implies that the right null space of X is non-trivial. That is, there exists a non-zero vector $q = (q_1, \dots, q_n)^T \in k^n$ such that

$$X \begin{pmatrix} q_1 \\ \vdots \\ q_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Now, observe that the element $1_A \in A$ maps under Ψ to $(v_1, \dots, v_n) \in U$. Because $(v_1, \dots, v_n) \in U$, there exists some $(u_1, \dots, u_r) \in V^{\oplus r}$ such that

$$(v_1, \dots, v_n) = (u_1, \dots, u_r)X.$$

Multiplying both sides on the right by the column vector q :

$$\sum_{i=1}^n q_i v_i = (v_1, \dots, v_n) \begin{pmatrix} q_1 \\ \vdots \\ q_n \end{pmatrix} = (u_1, \dots, u_r) X \begin{pmatrix} q_1 \\ \vdots \\ q_n \end{pmatrix} = (u_1, \dots, u_r) \cdot 0 = 0.$$

This produces a non-trivial linear combination $\sum_{i=1}^n q_i v_i = 0$ with not all $q_i = 0$. But this directly contradicts the hypothesis that $\{v_1, \dots, v_n\}$ is linearly independent!

Therefore, the assumption $r < n$ is false. We must have $r = n$, which forces $U = V^{\oplus n}$. Hence, the map Ψ is surjective, and there exists $a \in A$ such that $av_i = w_i$ for each $i \in \{1, \dots, n\}$. ■

Theorem 3.9 (The Density Theorem for Finite-Dimensional Algebras)

Let A be an algebra over an algebraically closed field k .

1. If V is a finite-dimensional simple A -module, the representation map

$$\rho: A \longrightarrow \text{End}_k(V)$$

is surjective.

2. If $V_1, V_2, \dots, V_r$ are pairwise non-isomorphic finite-dimensional simple A -modules, the direct sum representation map

$$\rho_1 \oplus \dots \oplus \rho_r: A \longrightarrow \bigoplus_{i=1}^r \text{End}_k(V_i)$$

is surjective.

Proof. **Part (1):** Let $d = \dim_k(V)$, and let $\{v_1, \dots, v_d\}$ be a vector space basis of

V . Let $C \in \text{End}_k(V)$ be any arbitrary linear operator on V . We want to find an element $a \in A$ such that $\rho(a) = C$.

Define the vectors $w_i = C(v_i) \in V$ for each $1 \leq i \leq d$. Since $\{v_1, \dots, v_d\}$ is linearly independent in V , Lemma 3.5 guarantees the existence of an element $a \in A$ such that

$$a \cdot v_i = w_i = C(v_i) \quad \text{for all } 1 \leq i \leq d.$$

Since a linear transformation is entirely determined by its action on a basis, $\rho(a)v = C(v)$ for all $v \in V$, meaning $\rho(a) = C$. Thus ρ is surjective.

Part (2): Let $B \subseteq \bigoplus_{i=1}^r \text{End}_k(V_i)$ be the image of A under $\bigoplus_{i=1}^r \rho_i$. Notice that each $\text{End}_k(V_i)$, regarded as an A -module via the action $a \cdot M = \rho_i(a)M$, is isomorphic to $V_i^{\oplus d_i}$ where $d_i = \dim_k(V_i)$ (since matrix multiplication acts independently on each of the d_i columns).

Because $V_1, \dots, V_r$ are simple and pairwise non-isomorphic, the ambient space $\bigoplus_{i=1}^r \text{End}_k(V_i)$ is a semisimple A -module with isotypic components $\text{End}_k(V_i) \cong V_i^{\oplus d_i}$.

The image $B \subseteq \bigoplus_{i=1}^r \text{End}_k(V_i)$ is an A -submodule. By semisimplicity, B decomposes as

$$B = \bigoplus_{i=1}^r (B \cap \text{End}_k(V_i)).$$

Let $\pi_i: \bigoplus_{j=1}^r \text{End}_k(V_j) \rightarrow \text{End}_k(V_i)$ be the canonical projection onto the i -th component. By Part (1), the projection $\pi_i(B) = \rho_i(A) = \text{End}_k(V_i)$ is surjective for every i . Since there are no non-zero A -homomorphisms between V_i and V_j for $i \neq j$ (by Schur's Lemma), any submodule B whose projections cover each factor must be the entire direct sum:

$$B = \bigoplus_{i=1}^r \text{End}_k(V_i).$$

Therefore, $\bigoplus_{i=1}^r \rho_i$ is surjective. ■

Corollary 3.10 (Artin-Wedderburn Structure Theorem)

Let A be a finite-dimensional algebra over an algebraically closed field k . Then:

1. A has only finitely many pairwise non-isomorphic simple modules $V_1, \dots, V_r$.
2. There is a canonical isomorphism of algebras:

$$A/\text{Rad}(A) \cong \bigoplus_{i=1}^r \text{End}_k(V_i) \cong \bigoplus_{i=1}^r M_{d_i}(k), \quad \text{where } d_i = \dim_k(V_i).$$

3. In particular, if A is semisimple ($\text{Rad}(A) = \{0\}$), then A is isomorphic to a direct sum of matrix algebras:

$$A \cong \bigoplus_{i=1}^r M_{d_i}(k).$$

Proof. Consider any collection of pairwise non-isomorphic simple A -modules $V_1, \dots, V_r$.

By Theorem 3.5, the map

$$\bigoplus_{i=1}^r \rho_i: A \longrightarrow \bigoplus_{i=1}^r \text{End}_k(V_i)$$

is surjective. Its kernel is $\bigcap_{i=1}^r \ker(\rho_i) = \text{Rad}(A)$. By the First Isomorphism Theorem for algebras:

$$A/\text{Rad}(A) \cong \bigoplus_{i=1}^r \text{End}_k(V_i) \cong \bigoplus_{i=1}^r M_{d_i}(k).$$

Taking dimensions over k :

$$\dim_k(A) \geq \dim_k(A/\text{Rad}(A)) = \sum_{i=1}^r d_i^2.$$

Since each $d_i \geq 1$ and $\dim_k(A) < \infty$, the number of pairwise non-isomorphic simple modules r is strictly bounded by $\dim_k(A)$:

$$r \leq \sum_{i=1}^r d_i^2 \leq \dim_k(A) < \infty.$$

Thus, there can only be finitely many simple modules. If A is semisimple, $\text{Rad}(A) = \{0\}$, immediately yielding $A \cong \bigoplus_{i=1}^r M_{d_i}(k)$. ■

3.6 Maschke's Theorem

Having established the structure of semisimple algebras, we now address the cornerstone theorem that connects group theory to algebra semisimplicity: Maschke's Theorem.

Theorem 3.11 (Maschke's Theorem)

Let G be a finite group, and let k be a field whose characteristic does not divide the order of G (i.e., $\text{char}(k) \nmid |G|$). Then every finite-dimensional representation of G over k is completely reducible (semisimple).

In particular, the group algebra $k[G]$ is a semisimple algebra.

Proof. Let V be a finite-dimensional representation of G over k , and let $W \subseteq V$ be a G -invariant subspace (subrepresentation). To prove that V is semisimple, it suffices to show that W admits a G -invariant complement W' , so that $V = W \oplus W'$.

Step 1: Choose a vector space projection.

By elementary linear algebra, choose any vector space complement U of W in V , so $V = W \oplus U$. Let $P: V \rightarrow W$ be the canonical vector space projection of V onto W along U . By definition:

- $P(v) \in W$ for all $v \in V$;
- $P(w) = w$ for all $w \in W$ (i.e., $P|_W = \text{id}_W$);
- $P^2 = P$.

However, this projection P is generally *not* a G -homomorphism; that is, $P(\rho(g)v) \neq \rho(g)P(v)$ in general.

Step 2: Construct the Reynolds Averaging Operator.

To construct a G -equivariant projection, we average the conjugates of P over the entire group G . Define the operator $\bar{P}: V \rightarrow V$ by:

$$\bar{P} = \frac{1}{|G|} \sum_{g \in G} \rho(g) \circ P \circ \rho(g^{-1}).$$

Here, the hypothesis $\text{char}(k) \nmid |G|$ is critical: it guarantees that $|G| \cdot 1_k \neq 0$ in the field k , making the scalar $\frac{1}{|G|} \in k$ well-defined and invertible.

Step 3: Verify that $\bar{P}$ is a G -homomorphism.

Let $h \in G$. We compute the conjugation of $\bar{P}$ by $\rho(h)$:

$$\begin{aligned} \rho(h) \circ \bar{P} \circ \rho(h^{-1}) &= \rho(h) \left(\frac{1}{|G|} \sum_{g \in G} \rho(g) \circ P \circ \rho(g^{-1}) \right) \rho(h^{-1}) \\ &= \frac{1}{|G|} \sum_{g \in G} \rho(h) \rho(g) \circ P \circ \rho(g^{-1}) \rho(h^{-1}) \\ &= \frac{1}{|G|} \sum_{g \in G} \rho(hg) \circ P \circ \rho((hg)^{-1}). \end{aligned}$$

As g ranges over all elements of G , the product $x = hg$ also ranges over all elements of G exactly once (since left translation by h is a bijection of G). Therefore:

$$\rho(h) \circ \bar{P} \circ \rho(h^{-1}) = \frac{1}{|G|} \sum_{x \in G} \rho(x) \circ P \circ \rho(x^{-1}) = \bar{P}.$$

Multiplying both sides on the right by $\rho(h)$ yields:

$$\rho(h) \circ \bar{P} = \bar{P} \circ \rho(h) \quad \text{for all } h \in G.$$

Thus, $\bar{P} \in \text{End}_G(V)$ is an intertwining operator.

Step 4: Verify that $\bar{P}$ is a projection onto W .

First, take any vector $v \in V$. For each $g \in G$, $P(\rho(g^{-1})v) \in W$ because the image of P is W . Because W is a G -invariant subspace, applying $\rho(g)$ preserves W :

$$\rho(g) (P(\rho(g^{-1})v)) \in W.$$

Thus, every term in the sum defining $\bar{P}(v)$ lies in W . Because W is a subspace closed under linear combinations, $\bar{P}(v) \in W$ for all $v \in V$. Hence $\text{im}(\bar{P}) \subseteq W$.

Next, take any vector $w \in W$. Since W is G -invariant, $\rho(g^{-1})w \in W$. Because $P|_W = \text{id}_W$, we have $P(\rho(g^{-1})w) = \rho(g^{-1})w$. Applying $\rho(g)$:

$$\rho(g) (P(\rho(g^{-1})w)) = \rho(g)\rho(g^{-1})w = \rho(e)w = w.$$

Substituting this into the definition of $\bar{P}$:

$$\bar{P}(w) = \frac{1}{|G|} \sum_{g \in G} w = \frac{1}{|G|} \cdot |G|w = w.$$

Therefore, $\bar{P}|_W = \text{id}_W$. Since $\text{im}(\bar{P}) \subseteq W$ and $\bar{P}(w) = w$ for all $w \in W$, it follows that $\bar{P}^2 = \bar{P}$ and $\text{im}(\bar{P}) = W$. Thus $\bar{P}$ is a projection onto W .

Step 5: Define the complementary subrepresentation.

Set $W' = \ker(\bar{P})$. Since $\bar{P}$ is a projection onto W , basic linear algebra implies

$$V = \text{im}(\bar{P}) \oplus \ker(\bar{P}) = W \oplus W'.$$

Finally, we show that W' is G -invariant. Let $w' \in W'$ and $g \in G$. Then $\bar{P}(w') = 0$. Using the intertwining property from Step 3:

$$\bar{P}(\rho(g)w') = \rho(g)\bar{P}(w') = \rho(g)(0) = 0.$$

Hence $\rho(g)w' \in \ker(\bar{P}) = W'$. Therefore, W' is a G -invariant subspace.

By induction on dimension, every finite-dimensional representation decomposes into a direct sum of irreducible representations. In particular, applying this to the left regular representation $V = k[G]$, the algebra $k[G]$ decomposes as a direct sum of simple left $k[G]$ -modules, which by definition means $k[G]$ is a semisimple algebra. ■

Corollary 3.12 (Decomposition of the Group Algebra)

Let G be a finite group and k an algebraically closed field such that $\text{char}(k) \nmid |G|$. Let $V_1, \dots, V_r$ be the complete set of pairwise non-isomorphic irreducible representations of G , and let $d_i = \dim_k(V_i)$. Then:

1. There is an algebra isomorphism:

$$k[G] \cong \bigoplus_{i=1}^r \text{End}_k(V_i) \cong \bigoplus_{i=1}^r M_{d_i}(k).$$

2. The order of the group equals the sum of the squares of the degrees of the irreducible representations:

$$|G| = \sum_{i=1}^r d_i^2 = d_1^2 + d_2^2 + \dots + d_r^2.$$

3. The regular representation decomposes as:

$$k[G] \cong \bigoplus_{i=1}^r d_i V_i = \bigoplus_{i=1}^r V_i^{\oplus d_i}.$$

That is, each irreducible representation V_i appears in the regular representation with multiplicity equal to its dimension d_i .

3.7 The Modular Case: When Maschke's Theorem Fails

In the lecture notes, considerable attention was dedicated to the question: "*Why does this fail when $\text{char}(k) \mid |G|$?*" The study of representations when the field characteristic divides the group order is called *modular representation theory*. In this regime, group algebras are generally **not** semisimple, and representations fail to be completely reducible.

Remark. The Canonical Counterexample: $G = \mathbb{Z}_2$ over $\mathbb{F}_2$.

Let $G = \mathbb{Z}_2 = \{e, g\}$ with $g^2 = e$, and let $k = \mathbb{F}_2 = \{0, 1\}$ be the finite field with two elements. Here $|G| = 2$, so $\text{char}(k) = 2$ divides $|G|$.

The regular representation $V = k[\mathbb{Z}_2]$ is a 2-dimensional vector space with basis $\{e, g\}$. The generator g acts on this basis by:

$$g \cdot e = g, \quad g \cdot g = e.$$

In the ordered basis (e, g) , the matrix representing g is

$$\rho(g) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Notice that over $\mathbb{F}_2$:

$$(\rho(g) - I)^2 = \rho(g)^2 - 2\rho(g) + I = I - 0 + I = 2I = 0.$$

Consider the 1-dimensional subspace:

$$W = \text{span}_{\mathbb{F}_2}\{e + g\}.$$

Since $g \cdot (e + g) = g + e = e + g$, W is a 1-dimensional G -invariant subrepresentation (isomorphic to the trivial representation).

Now, does W have a G -invariant complement W' in V ?

If W' were an invariant complement, $\dim_{\mathbb{F}_2}(W') = 1$, so $W' = \text{span}_{\mathbb{F}_2}\{v\}$ for some non-zero vector $v \notin W$. The remaining non-zero vectors in $\mathbb{F}_2^2$ are:

$$e \quad \text{and} \quad g.$$

However:

$$g \cdot e = g \notin \text{span}\{e\}, \quad g \cdot g = e \notin \text{span}\{g\}.$$

Neither $\text{span}\{e\}$ nor $\text{span}\{g\}$ is invariant under the action of g !

Indeed, any 1-dimensional invariant subspace must be spanned by an eigenvector of $\rho(g)$. The characteristic polynomial of $\rho(g)$ is

$$\det(tI - \rho(g)) = t^2 - 1 = (t - 1)^2 \in \mathbb{F}_2[t].$$

The eigenspace for $\lambda = 1$ is the null space of $\rho(g) - I = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, which is exactly $W = \text{span}\{e + g\}$, of dimension 1.

Changing to the basis $v_1 = e + g$ and $v_2 = e$, the matrix of $\rho(g)$ becomes:

$$P^{-1}\rho(g)P = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

which is a **single Jordan block of size 2!**

Consequently:

- V is *indecomposable*: it cannot be split into a direct sum $W \oplus W'$.
- V is *not irreducible*: it contains the proper subrepresentation W .
- Therefore, $k[\mathbb{Z}_2]$ is **not semisimple**.

The averaging operator fails because:

$$\sum_{h \in G} \rho(h)P\rho(h^{-1}) = P + \rho(g)P\rho(g),$$

which has trace zero in characteristic 2, and dividing by $|G| = 2 \equiv 0 \pmod{2}$ is undefined.

3.8 Concrete Examples and Applications

3.8.1 The Cyclic Group $\mathbb{Z}_2$ in Characteristic $\neq 2$

When $\text{char}(k) \neq 2$, the group algebra $k[\mathbb{Z}_2]$ decomposes according to Maschke's Theorem. Since $\mathbb{Z}_2$ is abelian, all irreducible representations are 1-dimensional. The condition $|G| = \sum d_i^2$ becomes $2 = 1^2 + 1^2$, giving exactly two non-isomorphic 1-dimensional representations:

1. **Trivial representation:** $\rho_{\text{triv}}(g) = [1]$.
2. **Sign representation:** $\rho_{\text{sgn}}(g) = [-1]$.

The idempotents that project onto these subrepresentations are:

$$e_{\text{triv}} = \frac{e + g}{2}, \quad e_{\text{sgn}} = \frac{e - g}{2}.$$

We readily check that $e_{\text{triv}}^2 = e_{\text{triv}}$, $e_{\text{sgn}}^2 = e_{\text{sgn}}$, and $e_{\text{triv}}e_{\text{sgn}} = 0$, giving the algebra isomorphism:

$$k[\mathbb{Z}_2] \cong k \oplus k.$$

3.8.2 The Symmetric Group S_3

The symmetric group on 3 elements, $S_3 = \langle \sigma, \tau \mid \sigma^3 = 1, \tau^2 = 1, \tau\sigma\tau = \sigma^{-1} \rangle$, has order $|S_3| = 6$.

Over $\mathbb{C}$, the dimension formula requires:

$$6 = \sum_{i=1}^r d_i^2.$$

The number of conjugacy classes of S_3 is 3 (corresponding to cycle types $(\cdot)(\cdot)(\cdot)$, $(\cdot\cdot)(\cdot)$, and $(\cdot\cdot\cdot)$), so $r = 3$. The unique positive integer solution to $6 = d_1^2 + d_2^2 + d_3^2$ is

$$6 = 1^2 + 1^2 + 2^2.$$

The three irreducible representations of S_3 are:

1. **Trivial Representation** V_1 ($d_1 = 1$): $\rho_{\text{triv}}(\pi) = 1$ for all $\pi \in S_3$.
2. **Sign Representation** V_2 ($d_2 = 1$): $\rho_{\text{sgn}}(\pi) = \text{sgn}(\pi) \in \{\pm 1\}$.
3. **Standard 2D Representation** V_3 ($d_3 = 2$):

Consider the natural permutation representation on k^3 with basis $\{e_1, e_2, e_3\}$.

The 1-dimensional subspace spanned by $e_1 + e_2 + e_3$ is invariant and carries the trivial representation. Its orthogonal complement

$$V_3 = \{(x_1, x_2, x_3) \in k^3 \mid x_1 + x_2 + x_3 = 0\}$$

is a 2-dimensional irreducible representation. Choosing a basis $u_1 = e_1 - e_2$ and $u_2 = e_2 - e_3$, the generators $\sigma = (1\ 2\ 3)$ and $\tau = (1\ 2)$ act by:

$$\rho_3(\sigma) = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad \rho_3(\tau) = \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix}.$$

By the Artin-Wedderburn Theorem:

$$\mathbb{C}[S_3] \cong \mathbb{C} \oplus \mathbb{C} \oplus M_2(\mathbb{C}).$$

The regular representation decomposes as:

$$\mathbb{C}[S_3] \cong V_{\text{triv}} \oplus V_{\text{sgn}} \oplus V_3^{\oplus 2},$$

illustrating that the 2-dimensional representation occurs with multiplicity 2 in the regular representation.

3.9 Exercises

Exercise 3.1 (The Center of the Group Algebra)

Let G be a finite group and k an algebraically closed field with $\text{char}(k) \nmid |G|$. Show that the dimension of the center of the group algebra $Z(k[G])$ equals the number of conjugacy classes of G , and deduce that the number of pairwise non-isomorphic irreducible representations of G equals the number of conjugacy classes of G .

Exercise 3.2 (Dual Representation)

Let $\rho: G \rightarrow \mathrm{GL}(V)$ be a representation of G . Define $\rho^*: G \rightarrow \mathrm{GL}(V^*)$ on the dual space $V^* = \mathrm{Hom}_k(V, k)$ by

$$(\rho^*(g)f)(v) = f(\rho(g^{-1})v) \quad \text{for } f \in V^*, v \in V.$$

Prove that ρ^* is a representation of G , and show that V^* is irreducible if and only if V is irreducible.

Chapter 4

Conclusion

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Appendix A

Appendix Title

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